

ANNAMALAI ARUNACHALAM

CTO & Chief Architect 

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PROFESSIONAL SUMMARY

Results-oriented technology leader with 28+ years spanning software engineering, solution architecture, product design, consulting, and IT management. For the last five years, CTO and Chief Architect of a privacy-first data-collaboration platform, owning the technology vision and end-to-end architecture (solution, cloud, data, blockchain, and Generative/Agentic AI). Proven at turning emerging technology (federated query, blockchain provenance, LLM and agentic systems) into shipped, revenue-generating products, while leading engineering teams and owning strategy, budgets, and delivery.

CORE COMPETENCIES

- **Technology strategy & architecture:** Set the technology vision and multi-year architecture roadmap; act as design authority across solution, cloud, data, blockchain, and AI, aligning technology with business goals.
- **Innovation & expertise:** Deep, hands-on capability across data management, blockchain, and Generative/Agentic AI (LLMs, agentic workflows, RAG, MCP), applied to enterprise-grade, cloud-native platforms.
- **Digital transformation:** Cloud computing, solution architecture, product design, system integration, and client services, with a recent focus on AI-first platforms.
- **Strategic & inspirational leadership:** Strategic planning, budget and P&L management, contract negotiation, resource optimization, and the ability to build and motivate high-performing teams.
- **Vendor & remote management:** Managing vendors, distributed development teams, and outsourcing across global locations without loss of delivery quality.

SKILLSET

- **Generative & Agentic AI:** LLMs (GPT, Claude, Gemini), RAG, NL2SQL, vector databases, prompt engineering, fine-tuning and transfer learning; agent frameworks and SDKs (CrewAI, Agno, PydanticAI, LangChain, Google ADK, Claude Agent SDK); Model Context Protocol (MCP), A2A protocol, multi-agent orchestration with local and hosted models.
- **Blockchain:** Smart-contract design and privacy-preserving data clean rooms using Rust, Go, Substrate, Hyperledger, Ethereum/Solidity, Node.js, IPFS, and Polygon.
- **Cloud computing:** AWS, Azure, and GCP - architecture, technology evaluation, DevOps, cost optimization, and cloud-neutral OSS migration.
- **Data management & analytics:** Federated/distributed query (Athena, Trino), RDS/PostgreSQL, DynamoDB/Couchbase, BigQuery, GCP Analytics Hub, Synapse Analytics, Data Factory; data partitioning and differential privacy.
- **Full-stack development:** Python and its AI/ML ecosystem (Pandas, NumPy, scikit-learn, SciPy, TensorFlow/Keras), with working command of Go, Rust, ReactJS, Node.js, .NET, VBA, and PL/SQL.

EXPERIENCE

Saptharushi

Jul 2021 - Present

CTO & Chief Architect

Lead all technology and product development for a privacy-first data-collaboration company. Own the technology vision and end-to-end architecture (solution, cloud, blockchain, data, and Generative/Agentic AI) for the flagship product, ATOM, and lead the engineering organisation, budgets, and delivery.

- **Data Clean Room (ATOM):** Designed and delivered a decentralized, privacy-compliant data clean room for MarTech and AdTech, enabling secure multi-party collaboration on first-party data to build advertising audiences without moving raw data or relying on third-party cookies.
- **Federated query engine:** Architected a distributed query platform where a central orchestrator splits SELECT-only, parameterized queries across private query servers deployed inside each data owner's own cloud (AWS, Azure, GCP). Sub-queries run locally where the data lives (predicate pushdown), and only hashed identifiers cross the trust boundary. Achieved a roughly 5-minute fetch across 250 million records spanning multiple client databases through parallel processing and hash-table partitioning.
- **Privacy & identity:** Deterministic SHA-256 hashed identifiers as the sole cross-dataset join key; PII hashing, encryption in transit and at rest, confidential computing (TEE), and differential privacy to prevent re-identification.
- **Blockchain provenance:** A Substrate (Parity) private, permissioned chain for immutable audit and provenance (GDPR Article 30), recording every query, activation, and deletion; snapshot hashes anchored on-chain (with IPFS backups) for integrity and dispute resolution. No cryptocurrency.
- **Microservices & platform:** Go microservices integrated through message queues, containerized on Kubernetes/EKS, with serverless components (Lambda) and Step Functions orchestration; passwordless Auth0 (email-OTP) with RBAC enforced at the API layer.
- **Generative AI (NL2SQL & RAG):** A natural-language querying interface integrated with the distributed engine - natural language converted to SQL by an LLM, validated, executed, and paraphrased back to the user with conversational memory. RAG pipelines generate insights over a client's private knowledge base via vector-DB similarity search.
- **Agentic AI prototyping:** Multi-agent applications using local and hosted LLMs with CrewAI, Agno, PydanticAI, LangChain, Google ADK, and the Claude Agent SDK; autonomous task planning and execution, with tool use exposed through MCP so agents can call data services on demand.
- **Consent management:** A blockchain-based consent solution with digital-wallet capabilities for consent traceability and compliance with GDPR and PDPA.
- **Cloud-neutral OSS migration:** Leading the migration from AWS-native services to open-source, cloud-neutral targets (Trino, Temporal, Couchbase, Valkey, vanilla Kubernetes, APISIX, OpenBAO, MinIO, and observability via Loki/OpenTelemetry).
- **Value-added services:** Conceptualized and designed activation, enrichment, analytics, insights, attribution, and settlement; activation across DV360, Meta, Google Ads, The Trade Desk, and others.
- **UX & multi-tenancy:** Designed the application UX and navigation, with configurable menus, logos, themes, and layouts enabling per-client look and feel.
- **Technology vision & strategy:** Defined and own the multi-year technology strategy and architecture roadmap; set architecture standards and principles (privacy-by-design, security, data governance) and communicate direction to founders, board, and stakeholders.
- **Organisation & R&D leadership:** Built and lead the engineering organisation (hiring, performance, culture, mentorship on Rust, Go, and Substrate); run technology evaluation, build-vs-buy decisions, and an R&D agenda (Agentic AI, MCP, cloud-neutral OSS) beyond immediate product needs.
- **Commercial & delivery leadership:** Represent technology in client, pre-sales, and partner engagements (RFP/RFI, demos, technical pitches); own the IT budget and vendor strategy. Delivered ATOM on schedule and within 20% of budget, saving SGD 2M through strategic resource optimization.

Sembcorp Industries

Oct 2019 - Jun 2021

Head of Digital Applications

- **CTRM/ETRM implementation:** Conducted business analysis to identify key pain points in trading and risk management, leading to the deployment of CTRM products (Lacima, Entrade) that improved efficiency by 25% and reduced manpower costs by 50%.
- **Trade configuration & management:** Configured physical trades with inventory capability and logistics management, ensuring accurate accounting transactions and compliance with company standards.
- **Risk management & reporting:** Managed physical and paper transactions, exporting data to the Lacima system as JSON for risk evaluation using parametric VaR and Monte Carlo simulation, producing

comprehensive market-exposure reports (volumetric, parametric VaR, Monte Carlo VaR, and spread exposure).

- **System integration & UI customization:** Integrated accounts with SAP via scheduled batch transfers, customized the user interface with real-time validations and alerts, and configured MFA and authorization for secure user access.
- **Credit & counterparty management:** Built a credit module to analyse exposure across counterparties for both physical and paper trades, integrated with an external legacy system to avoid redundancy and ensure data integrity.
- **Cloud & team leadership:** Planned and deployed the client/server workload to the cloud (IaaS and PaaS); hired, mentored, and managed support staff, achieving SGD 200k in savings via overseas remote hiring, a 15% increase in project profitability, and a 20% reduction in operational cost.

Mitsubishi

Apr 2015 - Sep 2019

Enterprise Solutions Architect

- **CTRM/ETRM deployment:** Led the successful implementation of the CTRM/ETRM system Aspect, improving trade-processing speed by 20% and reducing cost by 50%; evaluated solutions, negotiated costs, planned the program, and delivered in phases while managing budgets, remote resources, and vendor contracts.
- **Risk & reporting:** Developed and customized VaR modelling, MtM, PnL attribution, position exposure, credit exposure, and reconciliation reports (including Derivatives Settlement and Japanese tax-haven reporting); resolved major issues in the VaR and Credit modules with cost-effective solutions.
- **Automation & integration:** Implemented RPA (NTT Office Robot) for price-data processing, automated Aspect-to-SAP journal entries, and designed a comprehensive Aspect system spanning Front, Middle, and Back office functionality.
- **Technology & IT management:** Managed IT outsourcing, contract negotiations, budget preparation, technology evaluation, and procurement; migrated on-premise servers to Azure, reducing maintenance cost by 30% and improving reliability by 25%.
- **Leadership & collaboration:** Conducted demos, analyzed business requirements, negotiated vendor contracts, and collaborated with senior management on IT strategy; effectively managed a diverse team across time zones, fostering a cohesive working environment.

Aspect Enterprise Solutions

Oct 2010 - Mar 2015

Pre-sales Solutions Director

- **CTRM/ETRM presales:** Owned solution proposals, RFPs, RFQs, contract evaluations, account management, sales pitches, client presentations, demos, POCs, and pilot programs; closed 6 new clients and renewed SaaS contracts for 4 existing clients.
- **Client engagement & contract negotiation:** Conducted scoping workshops, developed proofs of concept and RFPs, negotiated contracts, and supported clients during program rollouts.
- **Project management & implementation:** Led implementation and project management, designing interfaces, business processes, and approval workflows using Aspect's proprietary scripting tool; developed custom queries and reports via the Direct Query Interface.
- **Product development:** Architected and developed the Forward Curve Builder (supporting inheritance, composite curves, and grid-point customization); defined product roadmaps and launched new modules for inventory and logistics management and trade templates.
- **Customer success & revenue protection:** Implemented CTRM solutions for 8+ customers, including complex projects for Chemoil, Concord Energy, Hanwa, Trafigura, MRI, and Itochu; resolved product issues that preserved a USD 3M revenue stream through a SaaS renewal.

OpenLink

Jan 2009 - Sep 2010

Senior Technical Consultant

- **CTRM/ETRM implementation:** Techno-functional lead implementing OpenLink Endur for Petronas across Front, Middle, and Back office, physical operations, inventory management, security, UI customization, custom reports, and notifications.

- **Risk management design:** Designed price indices and forward curves with custom logic for each price point; implemented business-process workflows to manage the trade lifecycle, integrating with Credit, VaR, and Exposure modules.
- **Workflow & reporting enhancements:** Modified workflows via TLM, developed a custom Position Exposure Report (APM), and configured VaR using RiskPak; delivered interfaces to legacy systems and third-party feeds (Reuters, Platts, TT, SWIFT) on schedule.
- **Cost savings & efficiency:** Developed custom reports in Oracle PL/SQL (stored procedures and materialized views) and delivered forward curves via Market Manager, avoiding the need for a new Forward Curve Builder and achieving significant cost savings.

Allegro

May 2005 - Dec 2008

Senior ETRM Implementation Consultant

- **Configuration & implementation:** Led the configuration of physical and paper trades, views, market pricing, forward curves, and risk modules; implemented custom weighted-average inventory valuation and designed comprehensive solutions.
- **Custom report development:** Designed stored procedures, materialized views, and database functions to support custom reports with client-specific business logic.
- **Data integration & uploaders:** Developed custom uploaders in .NET to process raw data files, convert them to system format, and store them for validation.
- **System & admin module design:** Designed Front office (derivatives and physicals), Middle office risk, Logistics/Operations, Inventory, and Back office settlement; built admin modules for security, alerts, reports, interfaces, workflows, and UI customization.
- **Delivery & cost savings:** Delivered customized CTRM solutions for PowerSeraya, SPC, Petro-Titan, Concord Energy, Pertamina, and Noble Group, achieving 30% cost savings via customized MTM PnL and volumetric analysis reports.

Energy Market Company

Feb 2003 - Apr 2005

Technical Consultant

- **System administration:** Managed installation, configuration, administration, and maintenance of WebLogic, SonicMQ, Solomon (accounting), OpenSSL (digital certificate authority server), and Oracle Database.
- **Market support:** Provided day-to-day support for market clearing, offer submission and processing, power scheduling, settlement processing, and market and settlement reruns.
- **Release management & training:** Formulated and launched release management and version-control procedures; trained users on offer submission, digital-certificate usage, and market reruns.
- **Analytics & compliance:** Developed operational-risk reports and RCAs for dispute resolution; designed a time-dimensional (star-schema) market-data model for analytics; participated in IT revamp for audit compliance and executed disaster-recovery exercises.

ISI-Dentsu

Jan 1998 - Jan 2003

Senior System Analyst

- **Systems integration services:** Led business requirements analysis, functional design, technical architecture, implementation, and maintenance, including data modelling and database maintenance using PowerBuilder, Oracle, Sybase, PL/SQL, SQLGupta, and ColdFusion.
- **Delivered enterprise systems:** Built a parts and purchase management system for JVC (PowerBuilder, Sybase); an accounting interface for Yakult (PowerBuilder, Oracle, batch integration); a loan management system for Sumitomo Bank (mainframe reflection interface, PowerBuilder, Oracle); and inventory management for Toyota Motors (SQLGupta, Oracle, PL/SQL).
- **Technology evaluation:** Evaluated ColdFusion and implemented a browser-based human-resource management application for Toyota Motors.

EDUCATION

- MBA, IT Strategic Management - Curtin University of Technology

- B.Eng, Computer Science and Engineering - University of Madras

COURSES & CERTIFICATIONS

- NUS Professional Certification - LLMs, Advanced NLP and Generative AI
- Coursera (INSEAD) - Blockchain Technologies
- Microsoft - Azure Solutions Architect Expert; Azure Data Engineer Associate
- Microsoft - Azure Fundamentals, Power Platform Fundamentals, Azure Data Fundamentals, Azure AI Fundamentals
- GCP - Digital Transformation, Innovating with Data, Cloud Identity, Architecting with GKE, BigQuery, Big Data and ML Fundamentals
- AWS - Cloud Practitioner; Well-Architected Framework
- Oracle DBA I and II, PL/SQL, Java, WebLogic, SonicMQ
- Product: Allegro, Endur/Findur, Aspect, Lacima RiskEngine, Entrade

RESIDENCY STATUS

Nationality - Singaporean.